2.12 Relation :

The notation of a relation between any two sets of objects is quite common. If we suppose A to be the set of all male human beings and B to be the set of all female human beings, then the relation F (father) can be defined between the sets A and B. Hence if $x \in A$ and $y \in B$, then we say x is related to y by the relation F if x is the father of y and write x F y. In our daily life, we often use the relations of the type "is the son of", "is the brother of", "is the husband of" etc. In Mathematics we generally use the relations "less than", "greather than", "less than or equal to", "greather than or equal to " etc.

Let A and B be any two non-empty sets. A relation R from A to B (usually called a binary relation) is a subset of $A \times B$. If $R \subseteq A \times B$ and $(a, b) \in R$, then we say that a is related to b by R and write a R b. If a is not related to b by R, then we write $a \not R b$.

If R is a relation from A to B, then R is a set of ordered pairs where the first element belongs to A and the second element belongs to B.

If R is a relation from a set A to itself i.e. R is a subset of $A^2 = A \times A$ ($R \subseteq A \times A$), then we say that R is a relation on A, instead of R is a relation from A to A.

2.13 Domain of a Relation :

The domain of a relation R is defined as the set of all first elements of the ordered pairs which belong to R.

Range of a Relation :

The range of a relation R is defined as the set of all second elements of the ordered pairs which belong to R.

Scanned by TapScanner

Type of Relations :

(i) Reflexive Relation :

A relation R defined on a set A is said to be reflexive if a R a, $\forall$ a $\in$ A.

i.e. (a, a) $\in$ R, $\forall$ a $\in$ A.

For example , let A = {x, y, z} and R = {(x, x), (y, y), (z, z)}, then R is a reflexive relation on A.

(ii) Symmetric Relation :

A relation R defined on a set A is said to be symmetric if a R b, then b R a,

$\forall$ a, b $\in$ A.

i.e. (a, b) $\in$ R $\Rightarrow$ (b, a) $\in$ R, $\forall$ a, b $\in$ A

For example, let A = {1, 2, 3} and R = {(1, 2), (1, 3), (2, 3), (3, 3), (3, 2), (2, 1), (3, 1)}.

Here (1, 2) $\in$ R $\Rightarrow$ (2, 1) $\in$ R, (1, 3) $\in$ R $\Rightarrow$ (3, 1) $\in$ R, (2, 3) $\in$ R $\Rightarrow$ (3, 2) $\in$ R

$\therefore$ R is a symmetric relation on A.

Scanned by TapScanner

(iii) Transitive Relation :

A relation R defined on a set A is said to be transitive if a R b, b R c,

then a R c, $\forall$ a, b, c $\in$ A.

i.e. (a, b) $\in$ R, (b, c) $\in$ R $\Rightarrow$ (a, c) $\in$ R, $\forall$ a, b,c $\in$ A

For example, let A = {a, b, c} and R = {(a, a), (b, b), (b, c), (c, a), (c, c), (b, a) }, then R is a transitive relation on A.

(iv) Anti-Symmetric Relation :

A relation R defined on a set A is said to be anti-symmetric if a R b, b R a,

then a = b, $\forall$ a, b $\in$ A.

i.e. (a, b) $\in$ R, (b, a) R $\Rightarrow$ a = b, $\forall$ a, b $\in$ A

For example, let N be the set of natural numbers and R be a relation on N defined by "a is a divisor of b" i.e. a R b if a divides b, then R is anti-symmetric because a divides b and b divides a $\Rightarrow$ a = b.

(v) Asymmetric Relation :

A relation R defined on a set A is said to be asymmetric if whenever a R b,

then b $\not$R a, $\forall$ a, b $\in$ A.

i.e. (a, b) $\in$ R but (b, a) $\notin$ R, $\forall$ a, b $\in$ A

For example, let A = {1, 2, 3} and R = { (1, 2), (1, 3), (3, 2)}, then R is an asymmetric relation on A.

(vi) Irreflexive Relation :

A relation R defined on a set A is said to be irreflexive if a $\not$R a, $\forall$ a $\in$ A.

i.e. (a, a) $\notin$ R, $\forall$ a $\in$ A

For example, let A = {a, b, c, d} and R = {(a, b), (b, c), (b, d), (c, a)}, then R is an irreflexive relation on A.

(vii) Compatible Relation :

A relation R defined on a set A is said to be compatible if it is both reflexive and symmetric.

i.e. (a, a) $\in$ R, $\forall$ a $\in$ A.

(a, b) $\in$ R $\Rightarrow$ (b, a) $\in$ R, $\forall$ a, b $\in$ A

(viii) Equivalence Relation :

A relation R defined on a set A is said to be equivalence relation if it is reflexive, symmetric and transitive.

Scanned by TapScanner

i.e. $(a, a) \in R, \forall a \in A$

$(a, b) \in R \Rightarrow (b, a) \in R, \forall a, b \in A$ and

$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R, \forall a, b, c \in A$

(ix) Universal Relation :

A relation R defined on a set A is said to be universal relation if $R = A \times A$

For example, let $A = \{a, b, c\}$, then

$R = \{(a, a), (a, b), (a, c), (b, a), (b, b), (b, c), (c, a), (c, b), (c, c)\}$ is a universal relation on A.

(x) Partial Order Relation : (Partially ordered set) POSET

A relation R defined on a set A is called a partial order relation if it is reflexive, anti-symmetric and transitive.

i.e. $(a, a) \in R, \forall a \in A$

$(a, b) \in R, (b, a) \in R \Rightarrow a = b, \forall a, b \in A$

$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R, \forall a, b, c \in A$

A set 'A' together with a partial order relation R is called partially ordered set or poset.

2.15 Transitive Extension :

Let R be a relation defined on a set A. Then a relation R_1 defined on the set A is called the transitive extension of R if $R \subseteq R_1$ and $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R_1$

For example, let $A = \{a, b, c, d\}$,

$R = \{(a, b), (b, c), (c, b), (b, d)\}$,

$R_1 = \{(a, b), (a, c), (a, d), (b, b), (b, c), (b, d), (c, b), (c, c), (c, d)\}$

Here $R \subseteq R_1$ and clearly $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R_1$

$\therefore$ R_1 is a transitive extension of R.

Transitive Closure :

Let R be a relation defined on a set A. Let R_1 denote the transitive extension of R, R_2 denote the transitive extension of $R_1, \ldots\ldots, R_{n+1}$ denote then transitive extension of R_n, then the transitive closure of R is defined as the union of

$R, R_1, R_2, R_3, \ldots\ldots, R_{n+1}, \ldots$ It is denoted by R^+.

$\therefore$ $R^+ = R \cup R_1 \cup R_2 \cup \ldots\ldots \cup R_n \cup R_{n+1} \cup \ldots\ldots$

Scanned by TapScanner

2.16 Examples on Relations

Example (7) : If both the relations A and B are reflexive, symmetric and transitive, then prove that $A \cap B$ is also reflexive, symmetric and transitive.

Solution : (i) Since A and B are reflexive

$\therefore$ x A x, x B x

$\therefore$ x $(A \cap B)$ x = (x A x) $\cap$ (x B x)

$\therefore$ $A \cap B$ is reflexive.

(ii) Since A and B are symmetric

$\therefore$ x A y $\Rightarrow$ y A x, and

x B y $\Rightarrow$ y B x

We now prove that x $(A \cap B)$ y $\Rightarrow$ y $(A \cap B)$x

x $(A \cap B)$ y = (x A y) $\cap$ (x B y)

$\Rightarrow$ (y A x) $\cap$ (y B x)

$\Rightarrow$ y $(A \cap B)$ x

$\therefore$ $A \cap B$ is symmetric

(iii) Since A and B are transitive

$\therefore$ x A y, y A z $\Rightarrow$ x A z and

x B y, y B z $\Rightarrow$ x B z

We now prove that x $(A \cap B)$ y, y $(A \cap B)$ z $\Rightarrow$ x $(A \cap B)$ z

x $(A \cap B)$ y, y $(A \cap B)$ z = (x A y) $\cap$ (x B y), (y A z) $\cap$ (y B z)

= (x A y), (y A z) $\cap$ (x B y), (y B z)

$\Rightarrow$ (x A z) $\cap$ (x B z)

$\Rightarrow$ x $(A \cap B)$ z

$\therefore$ $A \cap B$ is transitive

Example (8) : Prove that $R = \{x\ R\ y \mid x^2 - 4xy + 3y^2 = 0, \forall x, y \in N\}$ is reflexive, but not symmetric. (S-06)

Solution : (i) Reflexive :

We prove that $(x, x) \in R, \forall x \in N$

Now x R y = $x^2 - 4xy + 3y^2 = 0$

x R x = $x^2 - 4x^2 + 3x^2 = 0$. This is true

$\therefore$ R is reflexive

(ii) Symmetric :

We Verify whether x R y $\Rightarrow$ y R x

Scanned by TapScanner

We have $x R y = x^2 - 4xy + 3y^2 = 0$

Then $y R x = y^2 - 4xy + 3x^2 = 0.$ But this is not true

$\therefore$ R is not symmetric

Example (9) : If Z is a set of integers and a, b $\in$ Z, a R b if $b = a^r$ for some positive integer r, then show that R is a partial order on Z. **(S-06)**

Solution : **Given :** a R b if $b = a^r$

where r is a positive integer and a, b $\in$ Z

To prove : R is a partial order on Z

We prove that R is reflexive, antisymmetric and transitive.

(i) Reflexive : Let $r = 1$, $a = a^r = a^1 = a$

$\therefore$ a R a, $\forall$ a $\in$ Z

$\therefore$ R is reflexive.

(ii) Antisymmetric : Let a R b and b R a, $\forall$ a, b $\in$ Z

Then $b = a^r$ and $a = b^r$

$\Rightarrow$ $a = b$ (when $r = 1$)

i.e. a R b and b R a $\Rightarrow$ $a = b$ $\forall$ a, b $\in$ Z

$\therefore$ R is antisymmetric.

(iii)Transitive : Let a R b and b R c

Then $b = a^r$ and $c = b^r$

$c = (a^r)^r$ $(\because b = a^r)$

$c = a^{r^2}$

Since r is a positive integer, so is r^2

$\therefore$ a R c

i.e. a R b and b R c $\Rightarrow$ a R c

$\therefore$ R is transitive

$\therefore$ R is a partial order on Z.

Example (10) : Define a relation on the set S = {1, 2, 3, 4} which is (i) reflexive, symmetric but not transitive (ii) reflexive, transitive but not symmetric. **(S-06,11)**

Solution : **Given :** S = {1, 2, 3, 4}

(i) Let us define R as

R = { (1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (3, 4), (2, 4), (1, 4), (2, 1), (4, 1), (4, 3), (4, 2) }

Scanned by TapScanner

Clearly R is reflexive as $(a, a) \in R, \forall a \in S$

Also R is symmetric as $(a, b) \in R \Rightarrow (b, a) \in R, \forall a, b \in S$

But R is not transitive because $(3, 4) \in R, (4, 1) \in R$ but $(3, 1) \notin R$.

(ii) Let us define R as

$R = \{ (1, 1), (2, 2), (3, 3), (4, 4), (1, 2), (2, 3), (1, 3), (3, 4), (2, 4),$
$(2, 1), (3, 1), (1, 4) \}$

Clearly R is reflexive as $(a, a) \in R, \forall a \in S$

Also R is transitive as $(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R, \forall a, b, c, \in S$

But R is not symmetric because

$(3, 4) \in R$ but $(4, 3) \notin R$.

Example (11) : Show that the relation R defined on a set A is an equivalence relation.

(a, b) R (c, d) $\Leftrightarrow$ a = c (S-06)

Solution : **Given :** (a, b) R (c, d) $\Leftrightarrow$ a = c

To prove : R is an equivalence relation

We prove that R is reflexive, symmetric and transitive.

(i) Reflexive : We have (a, b) R (a, b) $\Leftrightarrow$ a = a

This holds true.

$\therefore$ R is reflexive.

(ii) Symmetric : We have (a, b) R (c, d)

$\Leftrightarrow$ a = c

$\Leftrightarrow$ c = a

$\Rightarrow$ (c, d) R (a, b)

$\therefore$ R is symmetric.

(iii)Transitive : We have (a, b) R (c, d) $\Leftrightarrow$ a = c

and (c, d) R (e, f) $\Leftrightarrow$ c = e

$\therefore$ (a, b) R (c, d) , (c, d) R (e, f) $\Leftrightarrow$ a = c and c= e.

$\Leftrightarrow$ a = e

$\Rightarrow$ (a, b) R (e, f)

$\therefore$ R is transitive.

$\therefore$ R is an equivalence relation.

Scanned by TapScanner

Example (18) : A relation R is defined by $R = \{(A, B)/A \subseteq B\}$. Show that R is a partial order relation. Is this relation compatible ? (W-08)

Soluiton : **Given :** $R = \{(A, B) \mid A \subseteq B\}$

To prove : R is a partial order relation.

We prove that R is reflexive, antisymmetric and transitive.

(i) Reflexive : We prove $A\ R\ A$

$A \subseteq A$ is trivially true

$\therefore$ $A\ R\ A$

$\therefore$ R is reflexive.

(ii) Antisymmetric : We prove $A\ R\ B$ and $B\ R\ A \Rightarrow A = B$

If $A \subseteq B$ and $B \subseteq A$, then $A = B$

$\therefore$ R is antisymmetric

(iii) Transitive : We prove $A\ R\ B$ and $B\ R\ C \Rightarrow A\ R\ C$

If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$

$\therefore$ R is transitive

$\therefore$ R is a partial order relation.

(iv) Symmetric : We verify whether $A\ R\ B \Rightarrow B\ R\ A$

If $A \subseteq B$, then $B \subseteq A$ need not be true

$\therefore$ R is not symmetric

$\therefore$ R is not a compatible relation.

Example (19) : Let R denote a relation on the set of ordered pairs of positive integers defined by (x, y) R (u, v) if and only if xv = yu, show that R is an equivalence relation. (S-11)

Solution : **Given :** $(x, y)\ R\ (u, v) \Leftrightarrow xv = yu$

To prove : R is an equivalence relation

We prove that R is reflexive, symmetric transitive.

(i) Reflexive : We prove $(x, y)\ R\ (x, y)$

Since $(x, y)\ R\ (u, v) \Leftrightarrow xv = yu$

$\therefore$ $(x, y)\ R\ (x, y) \Leftrightarrow xy = yx$

$\Leftrightarrow xy = xy$

This is trivially true

$\therefore$ R is reflexive.

Scanned by TapScanner

(ii) Symmetric : We prove $(x, y)\ R\ (u, v) \Rightarrow (u, v)\ R\ (x, y)$

Since $(x, y)\ R\ (u, v) \Leftrightarrow xv = yu$

$(u, v)\ R\ (x, y) \Leftrightarrow uy = vx$

$\Leftrightarrow yu = xv \Leftrightarrow xv = yu$

$\therefore$ $(x, y)\ R\ (u, v) \Rightarrow (u, v)\ R\ (x, y)$

$\therefore$ R is symmetric

(iii)Transitive : We prove $(x, y)\ R\ (u, v)$ and $(u, v)\ R\ (m, n) \Rightarrow (x, y)\ R\ (m, n)$

Since $(x, y)\ R\ (u, v) \Leftrightarrow xv = yu$

and $(u, v)\ R\ (m, n) \Leftrightarrow un = vm$

$\Leftrightarrow (xv)\ (un) = (yu)\ (vm)$

$\Leftrightarrow (xn)\ (vu) = (ym)\ (uv)$

$\Leftrightarrow (xn)\ (vu) = (ym)\ (vu)$

$\Leftrightarrow xn = ym$

$\Rightarrow (x, y)\ R\ (m, n)$

$\therefore$ $(x, y)\ R\ (u, v)$ and $(u, v)\ R\ (m, n) \Rightarrow (x, y)\ R\ (m, n)$

$\therefore$ R is transitive

$\therefore$ R is an equivalence relation.

Exampel (20) : Let $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ **and**

$R_1 = \{(x, y) \mid x - y$ **is divisible by 3**$\}$

$R_2 = \{(x, y) \mid x, y \in X$ **and x, y are in the same column of matrix A**$\}$

where $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ **(W-08,09, S-09)**

Find R_1 and R_2. Prove that R_1 and R_2 are equivalence relations. Also prove that $R_1 = R_2$.

Soluiton : **Given :** $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$R_1 = \{(x, y) \mid x - y$ is divisible by 3$\}$

$\therefore$ $R_1 = \{(1, 1), (1, 4), (1, 7), (2, 2), (2, 5), (2, 8), (3, 3), (3, 6), (3, 9),$

$(4, 1), (4, 4), (4, 7), (5,2), (5, 5), (5, 8), (6, 3), (6, 6), (6, 9), (7, 1), (7, 4),$

$(7, 7), (8, 2), (8, 5), (8, 8), (9, 3), (9, 6), (9, 9)\}$

Scanned by TapScanner

Given : $R_2 = \{(x, y) \mid x, y \in X$ and x, y are in the same column of matrix A$\}$

where $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

$\therefore$ $R_2 = \{(1, 1), (1, 4), (1, 7), (2, 2), (2, 5), (2, 8), (3, 3), (3, 6), (3, 9),$
$(4, 1), (4, 4), (4, 7), (5, 2), (5, 5), (5, 8), (6, 3), (6, 6), (6, 9),$
$(7, 1), (7, 4), (7, 7), (8, 2), (8, 5), (8, 8), (9, 3), (9, 6), (9, 9)\}$

Clearly $R_1 = R_2$

To prove : R_1 and R_2 are equivalence relations.

We prove that R_1 and R_2 are reflexive, symmetric and transitive.

(a) Reflexive : Consider R_1

Since $(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7), (8, 8), (9, 9) \in R_1$

$\therefore$ R_1 is reflexive

Since $R_1 = R_2$

$\therefore$ R_2 is also reflexive.

(b) Symmetric : Consider R_1

Clearly R_1 is symmetric because

$(a, b) \in R_1 \Rightarrow (b, a) \in R_1$ for all $a, b \in X$

Since $R_1 = R_2$

$\therefore$ R_2 is also symmetric

(c) Transitive : Consider R_1

Clearly R_1 is transitive because

$(a, b) \in R_1, (b,c) \in R_1 \Rightarrow (a,c) \in R_1$ for all $a, b,c \in X$

Since $R_1 = R_2$,

$\therefore$ R_2 is also transitive

$\therefore R_1$ and R_2 are equivalence relations.

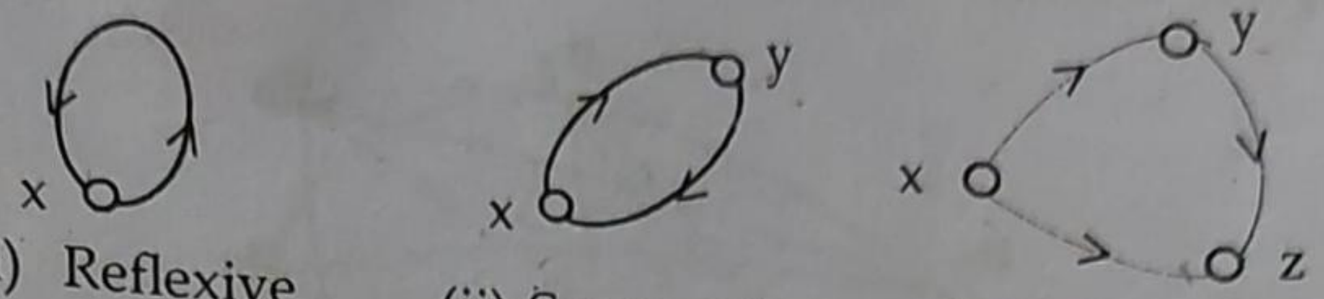
2.17 Relation Matrix :

Let R be a relation from one set X to another set Y. Then R can be represented by a matrix called relation matrix M_R and the elements of the relation matrix are given by

$$x_{ij} = \begin{cases} 1, \text{ if } x_i R y_j \\ 0, \text{ if } x_i \not R y_j \end{cases}$$

Scanned by TapScanner

Note : The graph of an equivalence relation can be represented as follows.



(i) Reflexive (ii) Symmetric (iii) Transitive

Fig.

Example (21) : Let $X = \{1, 2, 3, 4\}$ and $R = \{(x, y) \mid x > y\}$. Draw the graph of R and find the relation matrix M_R.

Solution : Given $X = \{1, 2, 3, 4\}$

$R = \{(x, y) \mid x > y\}$

$\therefore \quad R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2) (4, 3)\}$

Graph of R :

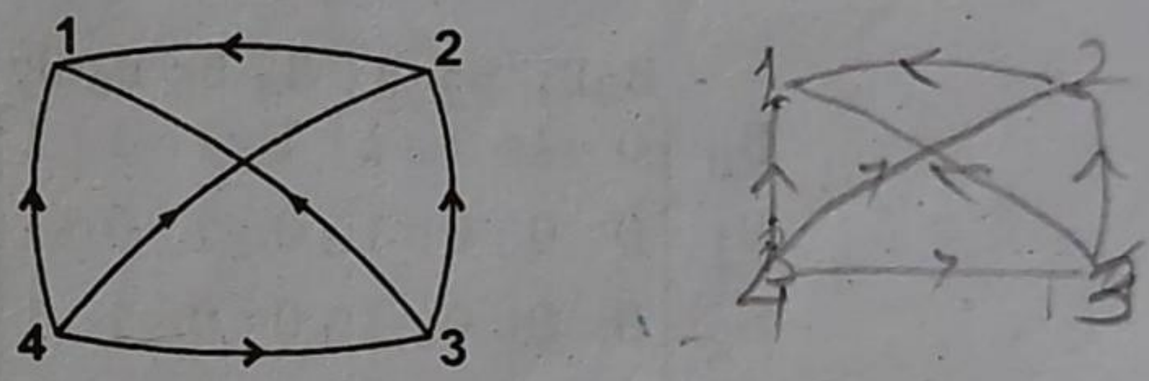


Fig.

Relation Matrix M_R :

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}.$$

Example (22) : Let $A = \{a, b, c\}$ and the subsets of A are denoted by $B_0 = \phi$, $B_1 = \{c\}$, $B_2 = \{b\}$, $B_3 = \{b, c\}$, $B_4 = \{a\}$, $B_5 = \{a, c\}$, $B_6 = \{a, b\}$, $B_7 = \{a, b, c\}$. If R is a relation of proper inclusion on the subsets B_0 to B_7, find the relation matrix M_R and draw the graph.

Soluiton : **Given :** $A = \{a, b, c\}$.

$B_0 = \phi$, $B_1 = \{c\}$, $B_2 = \{b\}$, $B_3 = \{b, c\}$, $B_4 = \{a\}$

$B_5 = \{a, c\}$, $B_6 = \{a, b\}$, $B_7 = \{a, b, c\}$.

Scanned by TapScanner

Graph of R is drawn below

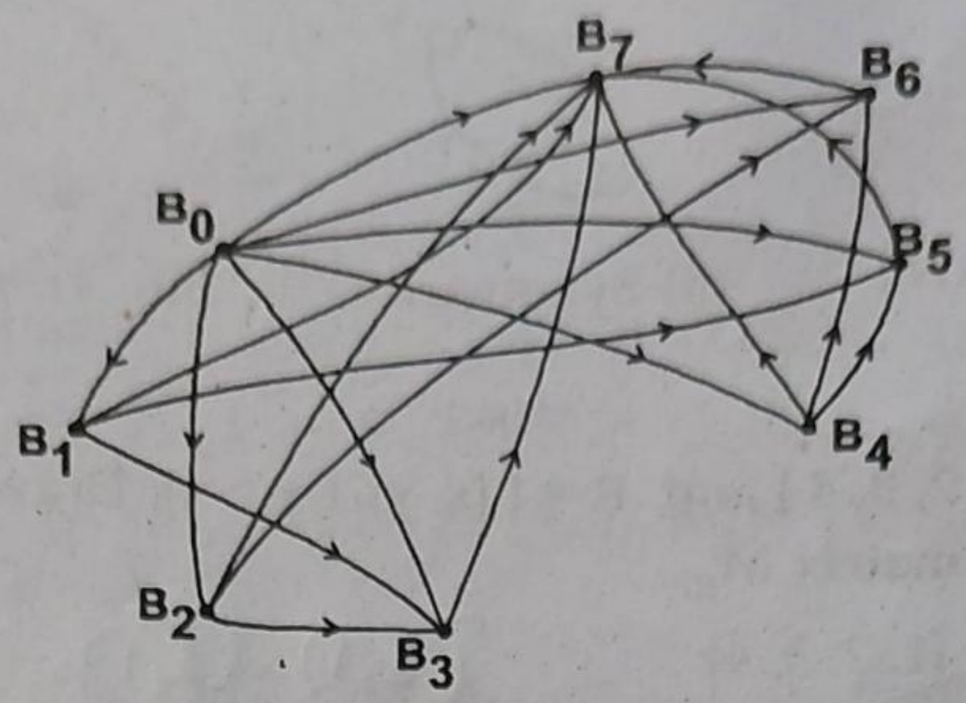


Fig.

Relation Matrix M_R :

$$M_R = \begin{array}{c} \\ B_0 \\ B_1 \\ B_2 \\ B_3 \\ B_4 \\ B_5 \\ B_6 \\ B_7 \end{array} \begin{array}{c} \begin{array}{cccccccc} B_0 & B_1 & B_2 & B_3 & B_4 & B_5 & B_6 & B_7 \end{array} \\ \left[\begin{array}{cccccccc} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

Example (23) : Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x, y) \mid x - y$ is divisible by 3$\}$. Draw the graph of R and find the relation matrix M_R. Show that R is an equivalence relation.

Soluiton : **Given :** $X = \{1, 2, 3, 4, 5, 6, 7\}$

$R = \{(x, y) \mid x - y$ is divisible by 3 $\}$

$\therefore$ $R = \{(1, 1), (1, 4), (1, 7), (2, 2), (2, 5), (3, 3), (3, 6), (4, 1), (4, 4), (4, 7), (5, 2), (5, 5), (6, 3), (6, 6), (6, 3), (7, 1), (7, 4), (7, 7)\}$

Scanned by TapScanner

Graph of R :

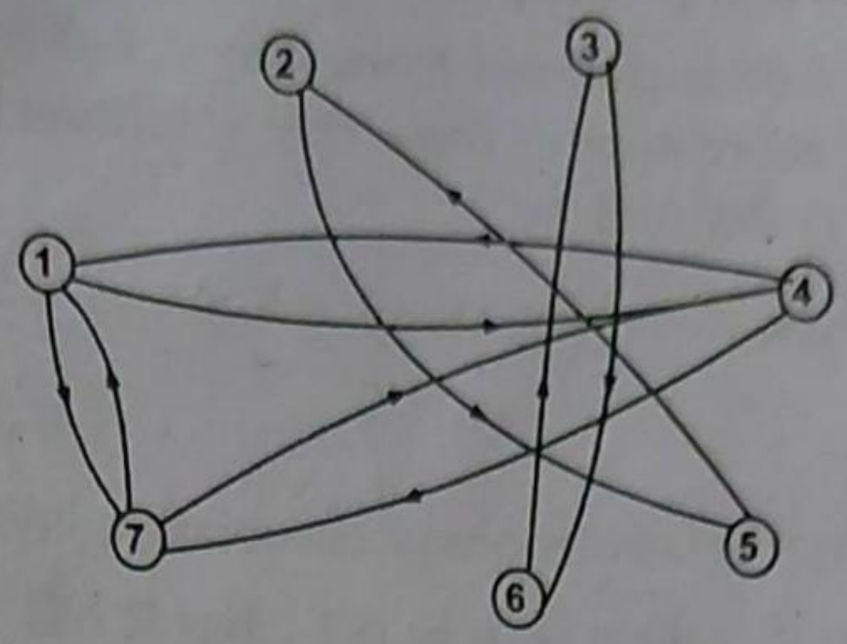


Fig.

Relation Matrix M_R :

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{array} \begin{array}{c} \begin{array}{ccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\ \left[\begin{array}{ccccccc} 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \end{array}$$

To prove : R is an equivalence relation.

We prove that R is reflexive, symmetric, transitive.

(a) Reflexive : We observe that

$(1, 1), (2, 2), (3, 3), (4, 4), (5, 5), (6, 6), (7, 7) \in R$

$\therefore$ R is reflexive

(b) Symmetric : We observe that

$(a, b)\ R \Rightarrow (b, a)\ R$, for all $a, b \in X$

$\therefore$ R is symmetric

(c) Transitive : We observe that

$(a, b) \in R, (b, c) \in R \Rightarrow (a, c) \in R$, for all $a, b, c \in X$

$\therefore$ R is transitive

$\therefore$ R is equivalence relation.

Scanned by TapScanner

2.18 Composite Relation (Composition of Relations) :

Let R be a relation defined from x to y. Let S be another relation defined from y to z. Then the composition of these two relations R and S is denoted by R o S (read as R circle S) and is defined by

$$RoS = \{(x, z) \mid \forall y, (x, y) \in R, (y, z) \in S\}$$

Note :

(1) Composition of relations obeys associative law, but does not obey commutative law, i.e. $R \circ (S \circ T) = (R \circ S) \circ T$, but $R \circ S \neq S \circ R$

(2) $M_{RoS} = M_R \cdot M_S$ or $M_{SoR} = M_S \cdot M_R$

(3) $M_{R^{-1}}$ = Transpose of M_R

(4) $M_{(RoS)^{-1}} = M_{S^{-1}} \cdot M_{R^{-1}}$

(5) $(M_R)'$ = Complement of M_R

Example (24) : Let R = { (1, 2), (3, 4), (2, 2)} and
S = { (2, 5), (4, 2), (3, 1), (1, 3) }, Find
(i)RoS (ii) SoR (iii)Ro(SoR) (iv)(RoS)oR (v)RoR (vi)SoS (vii) RoRoR

Solution : **Given :** R = {(1,2), (3,4), (2,2)}
S = {(2,5), (4,2), (3,1) (1,3)}

(i) RoS = {(1,5), (3,2), (2,5)}

(ii) SoR = {(4,2), (3,2,), (1,4)}

(iii) Ro (SoR) = {(3,2)}

(iv) (RoS) oR = {(3,2)}

(v) RoR = {(1,2)}

(vi) SoS = {(4,5),(3,3)}

(vii) RoRoR = {(1,2)}

Example (25) : Let R = {(1,2), (3,4), (2,2)} and
S = {(4,2), (2,5), (3,1), (1,3)} Find the relation matrix for R,S, RoS,SoR.

Solution : **Given :** R = {(1,2), (3,4), (2,1)}
S = {(4,2,),(2,5), (3,1), (1,3)}
Let X = {1,2,3,4,5}
RoS = {(1,5), (3,2), (2,5)}
SoR = {(4,2), (3,2), (1,4)}

Scanned by TapScanner

Relation Matrix for R (M_R) :

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Relation Matrix for S (M_S) :

$$M_S = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Relation Matrix for RoS (M_{RoS}) :

$$M_{RoS} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Relation Matrix for SoR (M_{SoR}) :

$$M_{SoR} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Scanned by TapScanner

2.19 Converse of a Relation :

Let R be a relation defined from X to Y . Then a relation defined from Y to X is called converse of relation R and is denoted by $\overline{R}$.

$$\overline{R} = \{(x,y) \mid (y,x) \in R\} = R^{-1}$$

If $x \in X, y \in Y$ then $x\,R\,y \Leftrightarrow y\,\overline{R}\,x$

Note :

$M_{\overline{R}} = M_{R^{-1}}$ = Transpose of M_R .

Example (26) : The relation matrices of two relations R and S are given by

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, M_S = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Find R, S, RoS and show that RoS is not an equivalence relation.

Soluiton : **Given :** $M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, M_S = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

Let $X = \{1,2,3\}$

Here $R = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$

$S = \{(1,1), (1,2), (2,1), (2,2), (2,3), (3,2), (3,3)\}$

$\therefore$ RoS $= \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,2), (3,3)\}$

$$M_{RoS} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

Now we prove that RoS is not an equivalence relation

Since $(1,1), (2,2), (3,3) \in$ RoS

$\therefore$ RoS is reflexive

Also $(1,3) \in$ RoS , but $(3,1) \notin$ RoS

$\therefore$ RoS is not symmetric

$\therefore$ RoS is not an equivalence relation.

Scanned by TapScanner